Novel Deep Learning Architectures for Brain Tumor MRI

SAETCN classification + SAS-Net segmentation

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Scope. The preprint proposes two custom deep-learning architectures: **SAETCN** for classification of glioma, meningioma, pituitary tumor and no-tumor MRI, and **SAS-Net** for brain-tumor segmentation. The reported headline results are 99.38% classification accuracy on one validation/test setting and 99.23% segmentation pixel accuracy.

1. Classification problem and datasets

Dataset	Composition	Reported split
Dataset 1 - Brain MRI/Kaggle	7,023 images; glioma 1,621, meningioma 1,645, pituitary 1,757, no tumor 2,000	5,712 train / 1,311 test
Dataset 2 - Figshare	3,064 T1 CE-MRI slices from 233 patients; 3 tumor classes	2,451 train / 613 test
Dataset 3 - custom cross-mixed	4,549 images; four classes	3,639 train / 910 test
Segmentation - BraTS 2020	Multimodal glioma MRI with expert tumor-region labels	Used for SAS-Net segmentation

Images are resized to 224 x 224. The paper describes contrast enhancement and normalization; Figshare MAT files are converted to image files and additionally normalized/rescaled. Classification uses Adam with learning rate 1e-4 and cross-entropy loss; segmentation uses BCE-with-logits loss.

Caution: this document is explicitly labeled a preprint and states that it had not been submitted for peer-reviewed journal publication at the time of the uploaded version.

2. SAETCN classification architecture

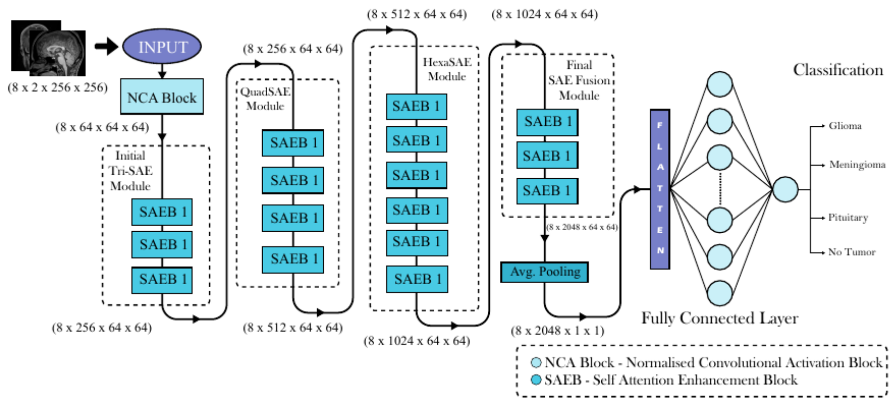


Figure 3: Self-Attention Enhanced Tumor Classification Network (SAETCN)

an improved residual network (ResNet) for brain tumor segmentation, addressing gradient diffusion issues and achieving higher precision compared to traditional methods. Their approach demonstrates the potential of enhanced deep learning models in improving segmentation accuracy. Bouhafra and El Bahi[5] reviewed deep learning approaches for brain tumor

Classification Network (SAETCN), designed to improve the accuracy of distinguishing between different types of brain tumors. SAETCN utilizes self-attention mechanisms to focus on relevant features within the medical imaging data, effectively capturing intricate patterns that differentiate tumor types. On the segmentation front, we introduce the Self-Attentive Seg-

Article Figure 3 (p. 4). SAETCN contains an initial normalized convolutional activation block followed by groups of custom Self-Attention Enhancement Blocks (SAEBs), adaptive average pooling and a fully connected classifier.

SAETCN contains 16 custom SAEB blocks arranged into Initial TriSAE, QuadSAE, HexaSAE and final SAE-fusion modules. Its design draws on residual skip connections and Inception-style parallel multiscale branches.

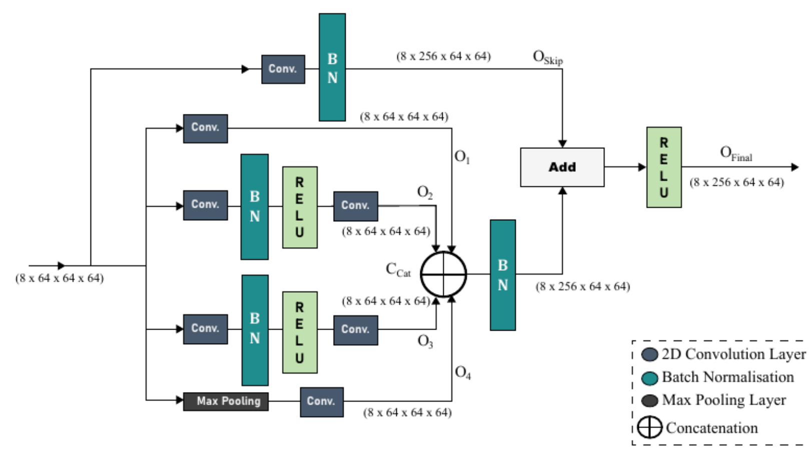


Figure 4: Self Attention Enhancement Block

Model. As shown in Figure 3 SAETCN consists of five main

internal operations of the K_{NCAB} is represented further in the

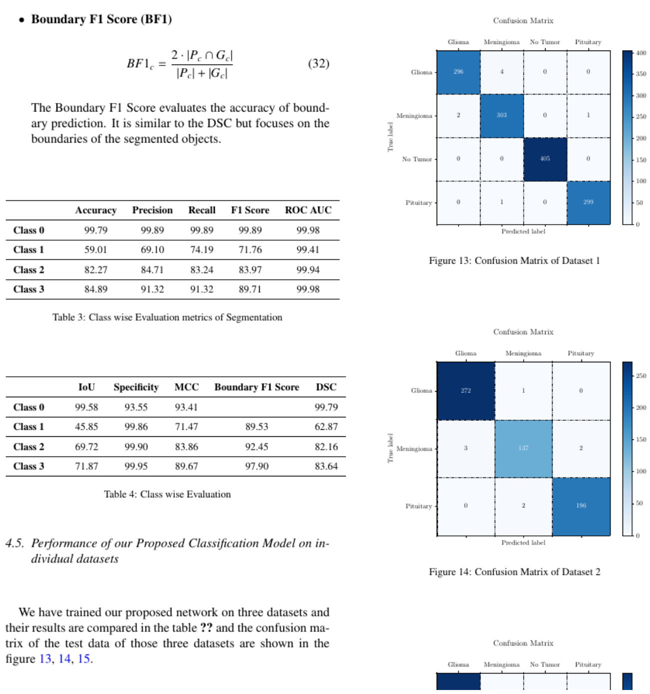
Article Figure 4 (p. 5). A SAEB splits the feature map into parallel convolution/pooling branches, concatenates their outputs, and combines the result with a skip projection before ReLU activation.

3. Why the SAEB design matters

The residual path is intended to improve gradient flow in a deep custom network. Parallel 1 x 1 / 3 x 3 / 5 x 5-style convolutional processing and pooling are intended to capture features at multiple spatial scales. Batch normalization and ReLU are used throughout. Despite the architecture's name, the paper's described SAEB mechanism is primarily a residual/Inception-style multibranch feature block rather than a conventional Transformer self-attention layer.

4. Classification results

Dataset	Accuracy	Precision	F1	Recall	ROC-AUC
Brain MRI / Dataset 1	99.38%	99.39%	99.39%	99.38%	99.99%
Figshare / Dataset 2	98.69%	98.69%	98.68%	98.69%	99.90%
Cross-mixed / Dataset 3	98.57%	98.59%	98.55%	98.57%	99.90%



- **Boundary F1 Score (BF1)**

$$BF1_c = \frac{2 \cdot |P_c \cap G_c|}{|P_c| + |G_c|} \quad (32)$$

The Boundary F1 Score evaluates the accuracy of boundary prediction. It is similar to the DSC but focuses on the boundaries of the segmented objects.

	Accuracy	Precision	Recall	F1 Score	ROC AUC
Class 0	99.79	99.89	99.89	99.89	99.98
Class 1	59.01	69.10	74.19	71.76	99.41
Class 2	82.27	84.71	83.24	83.97	99.94
Class 3	84.89	91.32	91.32	89.71	99.98

Table 3: Class wise Evaluation metrics of Segmentation

	IoU	Specificity	MCC	Boundary F1 Score	DSC
Class 0	99.58	93.55	93.41		99.79
Class 1	45.85	99.86	71.47	89.53	62.87
Class 2	69.72	99.90	83.86	92.45	82.16
Class 3	71.87	99.95	89.67	97.90	83.64

Table 4: Class wise Evaluation

4.5. Performance of our Proposed Classification Model on individual datasets

We have trained our proposed network on three datasets and their results are compared in the table **??** and the confusion matrix of the test data of those three datasets are shown in the figure 13, 14, 15.

Figure 13: Confusion Matrix of Dataset 1

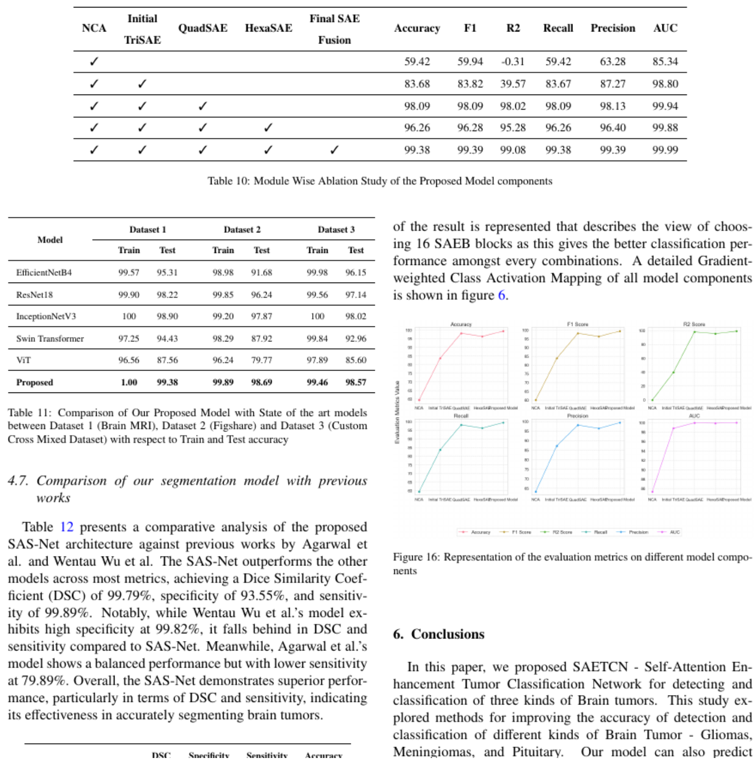
Figure 14: Confusion Matrix of Dataset 2

Article results page (p. 14). The paper presents confusion matrices and a cross-dataset metric table; Dataset 1 provides the strongest reported classification result.

The authors compare SAETCN against prior CNN and transfer-learning systems and argue that the custom architecture generalizes better across their three datasets. However, the paper itself acknowledges a remaining risk of overfitting and recommends training on substantially larger datasets before hospital deployment.

5. SAS-Net segmentation

SAS-Net is the companion segmentation network. It uses encoder/decoder processing and custom feature blocks to reconstruct a pixel-level tumor mask. The authors evaluate segmentation using pixel accuracy, Dice similarity coefficient, sensitivity, specificity, IoU and boundary-oriented measures.



NCA	Initial TriSAE	QuadSAE	HexaSAE	Final SAE Fusion	Accuracy	F1	R2	Recall	Precision	AUC
✓					59.42	59.94	-0.31	59.42	63.28	85.34
✓	✓				83.68	83.82	39.57	83.67	87.27	98.80
✓	✓	✓			98.09	98.09	98.02	98.09	98.13	99.94
✓	✓	✓	✓		96.26	96.28	95.28	96.26	96.40	99.88
✓	✓	✓	✓	✓	99.38	99.39	99.08	99.38	99.39	99.99

Table 10: Module Wise Ablation Study of the Proposed Model components

Model	Dataset 1		Dataset 2		Dataset 3	
	Train	Test	Train	Test	Train	Test
EfficientNetB4	99.57	95.31	98.98	91.68	99.98	96.15
ResNet18	99.90	98.22	99.85	96.24	99.56	97.14
InceptionNetV3	100	98.90	99.20	97.87	100	98.02
Swin Transformer	97.25	94.43	98.29	87.92	99.84	92.96
ViT	96.56	87.56	96.24	79.77	97.89	85.60
Proposed	**1.00**	**99.38**	**99.89**	**98.69**	**99.46**	**98.57**

Table 11: Comparison of Our Proposed Model with State of the art models between Dataset 1 (Brain MRI), Dataset 2 (Figshare) and Dataset 3 (Custom Cross Mixed Dataset) with respect to Train and Test accuracy

4.7. Comparison of our segmentation model with previous works

Table 12 presents a comparative analysis of the proposed SAS-Net architecture against previous works by Agarwal et al. and Wentau Wu et al. The SAS-Net outperforms the other models across most metrics, achieving a Dice Similarity Coefficient (DSC) of 99.79%, specificity of 93.55%, and sensitivity of 99.89%. Notably, while Wentau Wu et al.'s model exhibits high specificity at 99.82%, it falls behind in DSC and sensitivity compared to SAS-Net. Meanwhile, Agarwal et al.'s model shows a balanced performance but with lower sensitivity at 79.89%. Overall, the SAS-Net demonstrates superior performance, particularly in terms of DSC and sensitivity, indicating its effectiveness in accurately segmenting brain tumors.

	DSC	Specificity	Sensitivity	Accuracy

of the result is represented that describes the view of choosing 16 SAEB blocks as this gives the better classification performance amongst every combinations. A detailed Gradient-weighted Class Activation Mapping of all model components is shown in figure 6.

Figure 16: Representation of the evaluation metrics on different model components

6. Conclusions

In this paper, we proposed SAETCN - Self-Attention Enhancement Tumor Classification Network for detecting and classification of three kinds of Brain tumors. This study explored methods for improving the accuracy of detection and classification of different kinds of Brain Tumor - Gliomas, Meningiomas, and Pituitary. Our model can also predict

Article segmentation-results page (p. 16). The paper compares SAS-Net with two earlier segmentation methods and presents an ablation-style view of model-component contributions.

Method	DSC	Specificity	Sensitivity	Accuracy
Agarwal et al.	94.5%	92.6%	79.89%	91.3%
Wentau Wu et al.	89.58%	99.82%	91.10%	-
SAS-Net	99.79%	93.55%	99.89%	99.23%

6. Interpretation, strengths and limitations

The main strength is the attempt to solve **both classification and segmentation** within one research program using custom architectures rather than only fine-tuning an ImageNet backbone. The reported classification metrics are extremely high across three datasets, and SAS-Net reports very high Dice and sensitivity.

Those numbers should nevertheless be interpreted cautiously. The work is a preprint; Dataset 2 consists of multiple slices from 233 patients, while the paper describes image-level train/test counts rather than clearly documenting patient-level separation. If slices from the same patient were allowed across partitions, performance could be inflated by leakage. The paper also acknowledges possible overfitting and the need for larger-scale real-world validation.

The segmentation section reports very strong metrics, but clinical usefulness would require external testing across institutions/scanners, patient-level separation, uncertainty/calibration analysis and careful validation of boundary accuracy and failure cases.

7. Overall conclusion

SAETCN and SAS-Net illustrate a custom CNN strategy built around multibranch feature extraction, residual pathways and encoder-decoder segmentation. Within the datasets used by the authors, the architectures achieve very high reported performance. The most important next step is not simply further improvement in nominal accuracy, but demonstration that those results persist under strict patient-level splits and independent multi-institutional external validation.

The authors propose future expansion toward larger datasets, improved generalization, and mobile/web deployment that could support tumor detection and classification outside major clinical centers.